CN Practical 1: Survey Existing Network Infrastructure

Aim: To visit the computer laboratory, identify the type of network topology, connecting devices with their specifications, types of cables, network applications, and draw the layout of the installed network.

Theory:

A computer network is a collection of interconnected computers and other devices that communicate with each other to share information, hardware, software and other resources. Networks are widely used in computer laboratories, offices, colleges and industries. They make it possible to share printers, files, internet connections and application services.

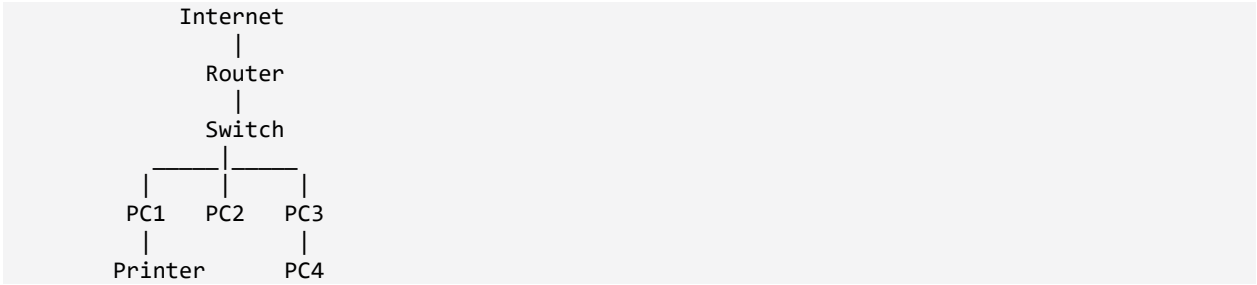
Network topology refers to the physical or logical arrangement of computers and networking devices. Common topologies include star, bus, ring, mesh and tree. In a star topology, all computers connect to a central switch or hub. It is commonly used in modern computer laboratories because it is easy to install, manage and troubleshoot.

Different networking devices perform different functions. A switch connects devices within a LAN, a router connects different networks, a modem provides access to an internet service, and a wireless access point connects wireless devices to a network. Network cables include twisted-pair cables, coaxial cables and fibre-optic cables. Twisted-pair Ethernet cables are commonly used to connect computers to switches.

Procedure:

1. Visit the computer laboratory and observe the installed network.
2. Identify the topology used to connect the computers.
3. Observe the networking devices, such as switches, routers and access points.
4. Note the specifications and purpose of each device.
5. Identify the network cables and their connectors.
6. Observe the applications used for internet access, file sharing and printing.
7. Determine how computers are connected to the central networking devices.
8. Draw a network layout showing computers, switches, routers and connections.

Network diagram:



Observation:

The laboratory network was observed and its topology, networking devices, cables and applications were identified. A central switch was found to connect multiple computers in the network. Ethernet cables were used for wired connectivity, and the network supported resource sharing and communication.

Conclusion:

The existing computer laboratory network infrastructure was studied successfully. Different network topologies, connecting devices, cable types and applications were identified. The layout of the installed network was drawn to understand the arrangement of the devices.